TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2014
Higher 2

CANDIDATE NAME						
CENTRE NUMBER	S					

INDEX NUMBER				

BIOLOGY 9648/03

Applications Paper and Planning Question — **Wednesday, 17 September 2014**

Paper 3 — **2 hours**

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.
The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show any working or if you do not use appropriate units.

At the end of the examination, fasten the answer papers securely and hand up separately.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's use	
1	**/14**
2	**/14**
3	**/12**
4	**/12**
5	**/20**
TOTAL	**/72**

This document consists of **17** printed pages and 1 blank page.

Answer **all** questions.

1 Fig. 1.1 illustrates a form of molecular cloning known as Sequence- and Ligation-Independent Cloning (SLIC). SLIC allows efficient and reproducible assembly of recombinant DNA with multiple gene inserts.

SLIC gene inserts can be generated using polymerase chain reaction (PCR) (Step1).

The SLIC procedure uses T4 DNA polymerase, which is also a 3' to 5' exonuclease, to generate single-stranded DNA overhangs in the insert and in the linearized vector fragment (Step 2). The resultant DNA overhangs can be about 16 to 18 nucleotides long and rich in G-C.

The gene insert and the linearized vector are then mixed together to form a recombination intermediate (Step 3).

Transformation is then carried out using competent *E. coli*, without the need for addition of DNA ligase (Step 4). Ligation is completed within the *E. coli*.

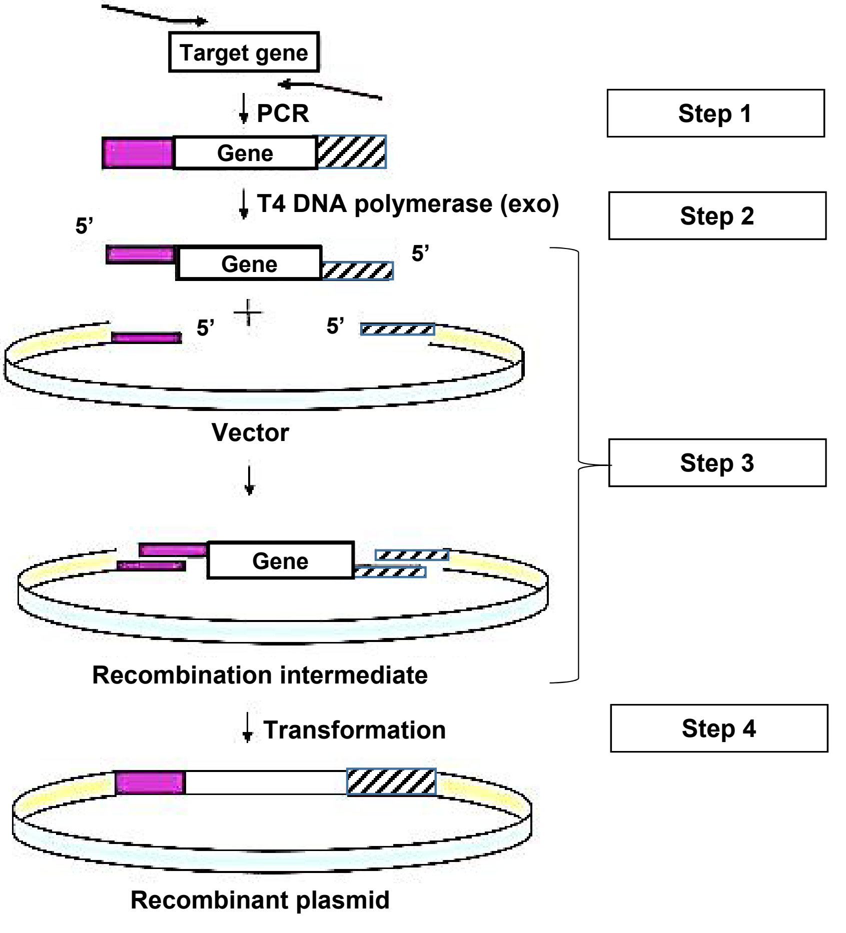


Fig. 1.1

(a) Describe how the target gene can be obtained from the organism. [2]

1. Isolate **mRNA** from **cells that actively express** the target gene
2. **Complementary DNA**/cDNA is synthesized using **reverse transcriptase**
3. **NaOH**/alkali added to **digest mRNA**
4. Add **DNA polymerase** to make cDNA **double stranded**

(b) With reference to **Step 3**,

(i) explain the advantage of SLIC compared to cloning using restriction endonucleases; [1]

1a. The target gene will be inserted in the correct orientation in the vector as the sticky ends on both sides of the gene are not complementary

1b. unlike cloning using restriction endonucleases which act on **palindromic sequences/** produces sticky ends on both sides of the gene are complementary / produces sticky ends on both sides of the gene (or vector) which can anneal/ OWTTE

(ii) suggest why the addition of DNA ligase is **not** required prior to transformation in SLIC. [1]

1. Resultant DNA overhangs are **rather long and rich in guanine and cytosine,** which base-pairs with **3 hydrogen bonds.**
2. Thus the **multiple** hydrogen bonds **hold the fragments together strongly**

(c) Explain what is meant by "competent *E. coli*". [1]

1. E. coli/ bacterial cells **capable of taking up DNA** from an **external solution**

(d) Outline the additional steps required if the objective of cloning is to harvest the protein products. [3]

1. **Identify the presence of gene of interest in transformed host cell**
2. **By replica plating and selection of the cells** that survive on the master plate but not on the replica plate.
3. This is followed by **gene probing by DNA hybridization** to identify the recombinant vectors containing the gene of interest.
4. **Grow the selected host cells in large quantities in a suitable medium.**
5. These bacteria can then be **induced to produce large amounts of the protein product.**
6. The protein product produced can be **extracted, purified and packaged.**

(e) Using a named example, explain the advantages of using bacteria to clone human genes. [3]

1. **Example – insulin/ human growth hormone/ somatotrophin/ antithrombin**
   Advantages (any 2):
2. Bacteria can be made to produce human protein **identical to that produced by human body.**
3. This reduces the need to acquire protein from **animal sources (bovine, porcine)** which can **cause severe allergies** or immune response
4. This **prevents contamination by disease-causing agents (e.g. virus) from animal sources.**
5. Overcomes **ethical objections** ***(compared to extracting protein from live animals)***

6. It also <u>reproduces rapidly</u>, doubling every half an hour under favourable conditions *(<u>Larger quantities</u> produced at a <u>cheaper</u> cost; <u>Faster</u>)*

Recently, it has been shown that *Agrobacterium tumefaciens* can transfer recombinant DNA into cultured human cells. The technique used is called protoplast fusion, where the cells are induced to fuse using polyethylene glycol (PEG). Fig. 1.2 shows the cells prior to fusion.

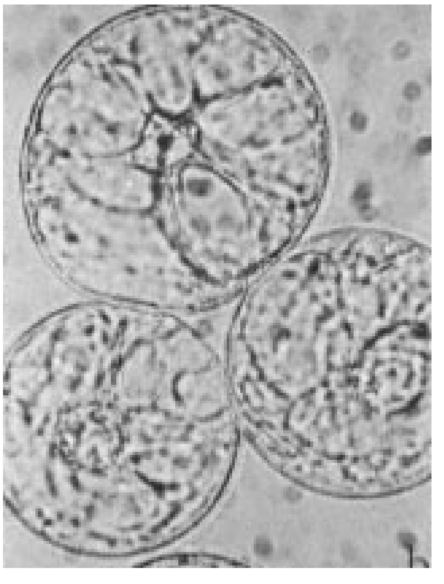


Fig. 1.2

(f) State how the Agrobacterium cells will need to be treated before protoplast fusion. [1]

1. **<u>Enzymes</u> (lysozyme) to <u>digest</u>** away the bacterial **<u>cell wall</u>**
   (Note: chloramphenicol added to amplify plasmid copy number)

(g) Outline **one** other method that can be used to transfer recombinant DNA into cultured human cells. [2]

1a. Using **<u>viruses</u>**
1b. A <u>region from the viral genetic material</u> can be replaced with the gene of interest.
1c. The recombinant genetic material is **<u>re-inserted into the viruses</u>.**
1d. Upon <u>infecting the host cells</u>, the viruses **<u>introduce</u>** their genetic material, including the gene of interest, into the infected cells.

OR
2a. <u>Microinjection</u>
2b. DNA injected into the target cell using **a <u>fine needle-like pipette</u>,** while viewing under a **<u>microscope</u>**. The cell is held steady by a **<u>larger pipette</u>**

OR
3a. Using <u>liposomes</u>
3b. Liposomes are small artificially **<u>created spheres (vesicles) surrounded by a phospholipid bilayer</u>** – similar to a membrane
3c. The <u>foreign DNA can be enclosed within</u> a liposome
3d. Upon incubation with the host cell, the liposomes may be **<u>taken into the cell via endocytosis</u>** and the DNA can be introduced into the cytoplasm

OR
4a. **Electroporation**
4b. The cells are stimulated with a **brief shock from a weak electric current**.
4c. this makes **holes appear transiently** in the **cell membrane**
4d. making them **permeable** to DNA and allowing the **DNA** molecules to **enter**.

OR
5a. **Gene gun**
5b. **DNA is coated onto minute gold particles.**
5c. These are **loaded into a gun** and once the trigger is pulled,
5d. a **low pressure helium pulse** delivers the DNA-coated gold particles into the cell.

[Total: 14]

2 Rice seeds lack the enzymes for two steps in the pathway for β-carotene production. The genes for these two enzymes have been inserted into rice embryos by genetic engineering, giving rise to Golden Rice™. This rice produces β-carotene in the endosperm of the seeds.

(a) Explain why there is a need to produce genetically modified rice which contains β-carotene. [2]

1. **People in developing countries suffer from vitamin A deficiency,**
2. **Rice is a staple diet in these countries**
3. **Golden rice provides people with β carotene which can form vitamin A / increases β-carotene in their diet / provide vitamin A in their diet**
4. **This can prevent blindness or reduces susceptibility to, diarrhoea, respiratory infections, measles and death.**

Fig. 2.1 shows the metabolic pathway by which β-carotene is synthesised in plants, and the enzymes that catalyse each step of the pathway.

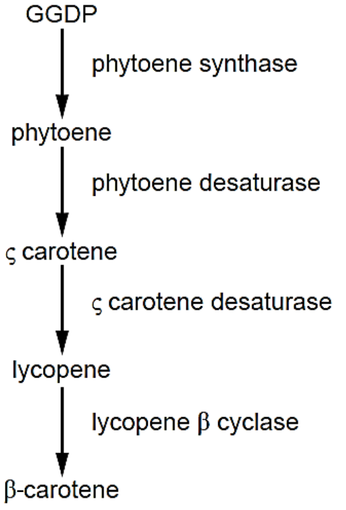


Fig. 2.1

The first types of Golden Rice™ contained a phytoene synthase gene, *psy*, from daffodils and a *crt* I gene, from the bacterium *Erwinia uredovora* which codes for the two desaturase enzymes.

Scientist noticed that these plants produced a very low mass of β-carotene per gram of rice. Measurements of the quantities of intermediates in the metabolic pathway in rice endosperm showed that there was always a large amount of GGDP present but no phytoene accumulated in the tissues.

(b) Explain how these results suggest it was **not** the enzymes produced by the *crt* I gene that was limiting the production of β-carotene. [1]

1. ***Crt* I gene codes for desaturase which catalyse the conversion of phytoene to lycopene**
2. **Phytoene synthase rather than desaturase which affect / catalyse the conversion of GDDP to phytoene**
   This explains for the large amount of GDDP in the cells.

Investigations were carried out to determine if *psy* genes taken from species other than daffodils would enable rice endosperm to produce greater quantities of β-carotene.

- *Psy* cDNA genes were obtained from maize, tomatoes, peppers and daffodils. The cDNA genes were inserted into different plasmids.
- Glu, a rice endosperm-specific glutelin promoter, and *crt* I genes from *E. uredovora*, were also inserted into all of the plasmids.
- The four types of genetically modified plasmids were then inserted into different cultures of rice cells.
- The quantity of β-carotene produced by these rice cells was measured.

  The results are shown in Table 2.1.

Table 2.1

Source of *psy* cDNA gene	**Total β-carotene of rice cells / arbitrary units**
Maize	14
Pepper	4
Tomato	6
Daffodil	1

(c) Explain how the *psy* cDNA gene was produced from each of the plant species. [2]

1. **mRNA is isolated from cells that actively express the *psy* gene**
2. **Complementary DNA (cDNA) copy is synthesized using the enzyme reverse transcriptase**
3. **Mixture treated with alkali / NaOH to digest mRNA**
4. **DNA polymerase is used to make the cDNA of the gene of interest double-stranded / synthesize the 2nd DNA strand.**

(d) Suggest why the Glu promoter was inserted into the plasmids. [1]

Any one:

1. **To ensure that the inserted genes will be expressed in the endosperm of the rice.**

2. **A suitable promoter may not be present in the endosperm of the rice.**

[Note:
The genes may not have been inserted in a correct orientation to the promoter in the nuclear DNA of rice cells leading to non-expression
This answer is applicable if the question is asking for a general reason]

(e) Explain whether the results in Table 2.1 support the hypothesis that the *psy* gene was limiting the production of β-carotene in genetically modified rice. [2]

1. **It supports / results support the hypothesis.**
2. ***Psy* gene from maize results in 14 a.u. of β-carotene in rice cells while *psy* gene from pepper results in 4 a.u. of β-carotene and *psy* gene from tomato results in 6 a.u. of β-carotene.**
3. **All rice cells contain the same *crt* I genes;**
4. **The only difference was the source of the psy genes ;**
5. If crtl gene was limiting there would be no difference in the carotene in each group. [bonus]

(f) The first type of Golden Rice™ was grown with the *psy* gene from daffodils because daffodils produce large amounts of β-carotene in their yellow petals, and they are monocotyledonous plants, like rice.

Suggest an explanation for the much lower production of β-carotene in rice containing the *psy* gene from daffodils than in rice containing the *psy* gene from maize. [2]

1. Psy gene from daffodils have slightly different base sequences compared to psy gene from maize.
➔ This results in a different 3-D conformation / shape of protein
➔ There is different interaction with factors involved in the biosynthesis, e.g. cofactors ;

OR

2. Psy gene from daffodils have slightly different base sequences compared to psy gene from maize.
➔ Affects binding of specific transcription factors
➔ Different rates of gene expression

OR

3. The cytoplasm in the rice cells is a different environment from daffodil
➔ polypeptide chain maybe folded differently
➔ Different 3-D conformation / shapes
➔ There is different interaction with factors involved in the biosynthesis, e.g. cofactors ;

The following steps were taken to produce Golden Rice™:

- **Step 1 -** A length of cDNA transgenes was made which includes a specific promoter + *psy* gene + *crt* I gene.
- **Step 2 -** Copies of this length of cDNA were inserted into Ti plasmids from the bacterium *Agrobacterium tumefaciens*.
- **Step 3 -** *A. tumefaciens* containing the Ti plasmids were mixed with rice embryos in tissue culture. The T-DNA with the transgenes were inserted into the nuclear DNA of rice embryos.
- **Step 4 -** The embryos were grown into calli, plantlets and then plants.

(g) Explain how plantlets can be formed from calli. [2]

1. **For <u>shoot formation,</u> <u>high amounts of cytokinin</u> and <u>low</u> amounts of <u>auxin</u> should be added**
2. **After successful establishment of the explant, <u>several microshoots</u> are produced within a few weeks.**
3. **After <u>microshoots</u> have reached an <u>appropriate length</u>, <u>transferred</u> to <u>root-inducing medium</u>**
4. **For <u>root formation</u>, <u>high</u> amounts of <u>auxin</u> and <u>low</u> amounts of <u>cytokinin</u> should be added**

(h) State **one** risk to the environment of growing genetically engineered crops and **one** possible risk to human health from eating these crops. [2]

Risk to environment:
1. **Gene transfer to wild-type relative or other related crops through cross-pollination. Changes to ecosystem through hydridisation.**
2. **Gene transfer from crop to crop via bacteria or viruses with unknown effect. Changes to ecosystem.**
3. **Recipient plant (plant that receives the gene) outcompetes natural variety or species / ref to 'superweed';;**
4. **Maybe toxic to non-pests;;**
5. **Unknown long term effects on the environment / biodiversity.**

Risk to humans:
1. **Allergy;;**
2. **Antibiotic resistance of gut bacteria if antibiotic marker used;;**
3. **Long term toxicity;;**
4. **Gene transfer from virus to human cells / gut bacteria, with unknown effect;;**
5. **No long term studies done on effects to human health.**

[Total: 14]

3 Gene therapy is used to treat the genetic disorder consisting of ADA deficiency. Affected individuals are unable to produce the enzyme adenosine deaminase (ADA). Without this enzyme, T lymphocytes cannot provide immunity to infection.

Fig. 3.1 shows the processes involved in the treatment of ADA deficiency by gene therapy.

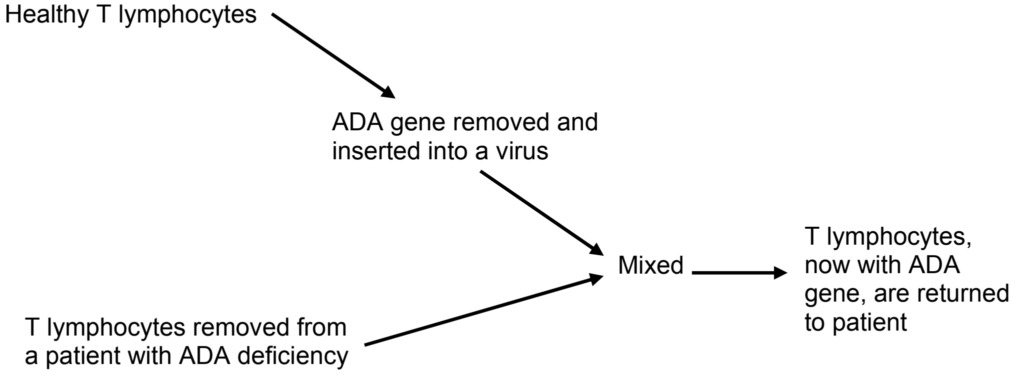


Fig. 3.1

(a) Explain what is meant by "gene therapy". [2]

1. **Transfer of a normal functional allele into the cells of the patient**
2. **to correct a genetic defect**
3. **via the use of viral or non-viral vectors**
4. **to restore normal phenotype / cure disease / normal phenotype can be expressed**

(b) The ADA gene is inserted into a virus. Give **two** advantages of using viral method as compared to a non-viral method in gene therapy. [2]

Any two:
1a. A viral gene delivery system allows for **targeting of specific cells** –
1b. **viral vectors** contain **surface proteins** that are **only recognised by receptors on** the **target cells** while the **non-viral** does **not** allow for **specific targeting of cells**. / The envelope protein of the viruses enables the viruses to infect human stem cells

2a. These lead to a **greater efficiency of gene transfer** as **compared to a non-viral** method.
2b. **Less vectors** are **needed** to ensure that **sufficient quantities** of the **normal functional allele enter the nucleus** of the stem cells / **incorporated** into the **host cell DNA** and expressed.

Comment [LY1]: These 2 points are actually related! Need to reorg

3. **Viral method allows for long-term expression of the normal allele as the alelle was first incorporated into the viral genome (can be DNA or RNA) which can be integrated into host cell DNA .**

(c) (i) Explain why the gene therapy method shown in Fig. 3.1 must be repeated at regular intervals for the treatment of ADA deficiency. [2]

1. **The T lymphocyte cell is fully differentiated / specialised cell.**

2. **The treated cell has a limited lifespan and do not divide.**

Therefore treatment has to be repeated to target new T lymphocytes.

(ii) Suggest a possible improvement to the process and explain how it will reduce the number of treatments given. [2]

1. **Target blood stem cells / haematopoietic stem cells.**

2a. **Stem cells have the ability to divide and self-renew, but lymphocytes do not.**

2b. **The genetically modified stem cells can persist for a longer period of time in the patient's body than genetically modified lymphocytes can.**

3a. **The genetically modified stem cells can differentiate and give rise to more lymphocytes with the functional allele.**

(d) Individuals who have been treated by this method of gene therapy do not pass the normal functional ADA allele to their children. Explain why. [1]

1. **Gene therapy treats somatic cells.**

2. **Germ cells (ie. Cells that give rise to gametes) / gametes are not treated and do not take up the normal functional allele.**
Hence, a person will still pass on the defective genes / do not pass normal functional allele to their children.

(e) Discuss the social and ethical concerns regarding the use of gene therapy. [3]

Any three:

1. **The use of viruses in gene present a variety of potential problems to the patients such as toxicity, immune and inflammatory responses.**

2. **There is also fear that the virus might mutate and gain virulence in the patient.**

3. **The insertion of the normal functional allele may disrupt the expression of other genes E.g. the random insertion of gene can cause the proto-oncogene to mutate to form an oncogene, leading to the development of tumors.**

OR

4. **Potential dangers and unpredictable factors with gene therapy makes its application risky. E.g. some SCID children treated with gene therapy developed cancer as a result.**

5. **It is difficult to determine the long-term effects of expo'sure to viral vectors and the effects these engineered viruses have on the human genome.**

6. **High cost. e.g. compared with conventional treatments. Only the wealthy and powerful will have access to therapies.**

7. **Altering one's genes goes against many religious beliefs. Some consider this as playing God.**

8. **Acceptable standards for gene therapy may be influenced by politicians, insurance companies and even physicians who may be driven by economical gains.**

9. **The decision of what is normal or disability may rest on just a small group of individuals or medical professionals. Some may consider disabilities as diseases which have to be cured or prevented.**
10. **Concerns that search for cure might demean the lives of individuals presently affected by disabilities or lead to new definitions of "normal".**

[Total: 12]

4 Planning Question

Enzymes catalysing essentially irreversible reactions are potential sites of control in cellular respiration. One of these enzymes is phosphofructokinase, which can be regulated by the reversible binding of citrate to its allosteric site. (Citrate is produced as an intermediate compound during Krebs cycle.)

Using this information and your own knowledge, design an experiment to determine the effect of citrate on the rate of cellular respiration.

You must use:

- 10 mM citrate,
- purified homogenate of enzymes found in the cytosol,
- 5% glucose solution,
- distilled water,
- benedict's solution,
- apparatus shown in Fig. 4.1, can be used to separate proteins from ions and disaccharides.

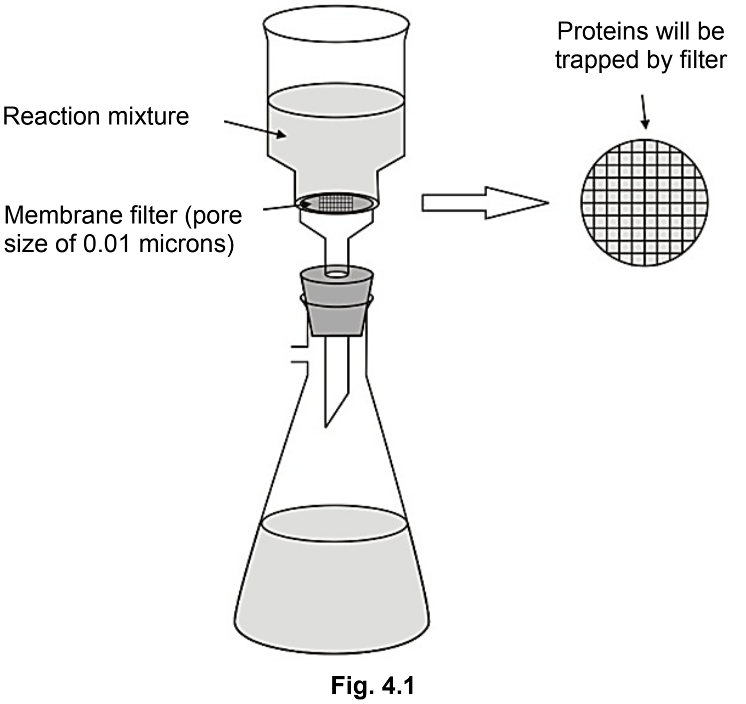


Fig. 4.1

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.,
- syringes,
- pipette fillers,
- white card,
- timer e.g. stopwatch or stop clock,
- thermometer,
- bunsen burner with tripod, gauze and bench mat.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment.

[Total: 12]

References:
http://www.ncbi.nlm.nih.gov/books/NBK22395/
http://www.ncbi.nlm.nih.gov/pmc/articles/PMC1489355/
http://www.nature.com/jid/journal/v59/n5/pdf/5617794a.pdf
http://elte.prompt.hu/sites/default/files/tananyagok/microbiology/images/4a76db6b.jpg
http://www.safewater.org/PDFS/resourcesknowthefacts/Ultrafiltration_Nano_ReverseOsm.pdf

ANSWER:
INTRODUCTION

A: RATIONALE

[Give a brief description of the experiment]

1. In this experiment, the effect of citrate on the rate of cellular respiration is determined by the rate of glucose consumption/amount of glucose remaining after a fixed period of time.

2. This is measured by comparing the results of the experiment against the results of the Benedict's test performed on a range of glucose solutions of known concentration/ standard glucose solutions.

[*Explain why this method is chosen*]

1. Citrate binds at the allosteric site of phosphofructokinase, which is one of the enzymes involved in glycolysis/ which catalyses the phosphorylation of fructose-6-phosphate to form fructose-1,6-bisphosphate.

2. Glucose is used as a substrate during glycolysis and its presence can be detected using the Benedict's test.

3. Glucose reacts with excess Benedict's solution to give precipitate. The colour and the cloudiness of the mixture reflect the amount of glucose present.

4. As the amount of glucose increases, the colour of the mixture changes from green to yellow to brown to brick-red and the degree of cloudiness increases.

   The concentration of the glucose can thus be estimated by comparing the colour and the degree of cloudiness against the mixture obtained from Benedict's test conducted on glucose solutions of known concentration (glucose standards).

METHODOLOGY

B: VARIABLES AND CONTROLLED VARIABLES

[*State the independent and dependent variables*]
The independent variable is citrate concentration.
The dependent variable is amount of glucose consumed per unit time.

[*Describe how the other variables will affect the results and how to keep them constant*]
Apart from citrate and glucose concentration, all other factors that can affect rate of cellular respiration should be kept constant as far as possible.

1. Initial concentration of glucose
   - It affects the rate of effective collision between enzymes and substrate and the rate of formation of enzyme-substrate complex.
   - Use a syringe to add the same volume of fresh glucose solution from the same stock (stirred well before use) to keep the initial concentration of glucose constant.

2. Temperature
   - It affects the kinetic energy of enzymes and substrate, thus changing the rate of effective collision between them.
   - The optimum temperature (37°C) is kept constant by using a thermostatically controlled water bath.

3. pH
   - It changes the conformation of the active site of the enzymes and lowers the rate of formation of enzyme-substrate complex.
   - Keep pH constant with a pH buffer.

4. Duration of reactions
   - It affects the amount of enzyme-substrate complexes that can be formed and hence the amount of products.
   - A digital stopwatch is used to ensure duration of reactions is kept constant.

C: DETAILED PROCEDURE

Note: All solutions are dispensed separately with clean and dry syringes and placed in clean and dry glassware (e.g. test tube and boiling tube) to prevent contamination or dilution.

Part 1: Preparation of the glucose standards

1. Label 10 boiling tubes 0.5%, 1.0%, 1.5%, 2.0%, 2.5%, 3.0%, 3.5%, 4.0%, 4.5% and 5.0%.

2. Prepare 20 ml of various concentrations of glucose solutions as shown in the table below. 10ml syringes are used to add the liquids and glucose solution is placed in their respective boiling tubes.

Concentration of glucose solution to be prepared /%	Volume of 5% glucose solution /ml	Volume of distilled water /ml
0.5	2	18
1.0	4	16
1.5	6	14
2.0	8	12
2.5	10	10
3.0	12	8
3.5	14	6
4.0	16	4
4.5	18	2

3. Label 10 test-tubes 0.5%, 1.0%, 1.5%, 2.0%, 2.5%, 3.0%, 3.5%, 4.0%, 4.5% and 5.0%.

4. Using a 2 ml syringe, add 0.5 ml of each concentration of glucose solution into their respective test-tubes.

5. Using a 5 ml syringe, add 5 ml of Benedict's solution. Shake gently to mix the contents of the tube.

6. Place the test-tubes in the boiling water for two minutes. Start the stopwatch.

7. After two minutes, stop the stopwatch. Remove the tubes from the boiling water and place them in a rack.

8. Shake gently to mix the contents of the tube and observe the contents of the test-tubes immediately after mixing. Record the observations in a table, noting any differences in terms of colour and cloudiness.

9. Set aside these tubes for comparison in Part 3.

Part 2: Preparation of citrate solution

10. Label 7 boiling tubes 0 mM, 1.5 mM, 3.0 mM, 4.5 mM, 6.0 mM, 7.5 mM, 9.0 mM.
    [At least 5 readings, regular intervals; maximum 10 mM]

11. Prepare 10 ml of various concentrations of citrate solution as shown in the table below. 10ml syringes are used to dispense the liquids and citrate solution into their respective boiling tube.

12. The boiling tubes are then placed in a thermostatically controlled water bath at 37°C for at least 10 minutes to equilibrate.

Concentration of citrate /mM	Amount of 10 mM citrate /ml	Amount of distilled water /ml
1.5	1.5	8.5
3.0	3	7
4.5	4.5	5.5

6.0	6	4
7.5	7.5	2.5
9.0	9	1

Part 3:

13. Label another 7 boiling-tubes 0 mM, 1.5 mM, 3.0 mM, 4.5 mM, 6.0 mM, 7.5 mM, 9.0 mM.

14. Using two clean 5 ml syringes, add 5 ml of **5% glucose solution [Note: concentration of glucose provided]** and 2 ml of pH buffer (e.g. 6.2) into a boiling-tube labeled 1.5 mM.

15. **Place this boiling tube and the purified homogenate of enzymes into a thermostatically controlled water bath at 37°C for 10 minutes to equilibrate. Start the stopwatch.

16. **After 10 minutes, stop the stopwatch. Using a 2ml syringe, add 2 ml of 1.5 mM citrate solution and using a 5 ml syringe, add 5 ml of enzyme homogenate into the boiling-tube. **[Note: enzyme homogenate should be the last to be added.]** Shake gently to mix the contents of the tube.

17. Place the test tube into a thermostatically controlled water bath of 37°C for 3 minutes. Start the stopwatch.

18. **After 3 minutes, quench the enzymatic reaction by placing the test tube in the boiling water bath for 2 minutes.

19. After 2 minutes, stop the stopwatch and pour the reaction mixture through the filtration set-up/membrane filter.

20. Perform Benedict's test on the filtrate as it was done for the glucose standards (steps 4-8).

21. Place the tubes containing the filtrate and the glucose standards against a white background. Compare with the glucose standards (from Part I, Step 9) to determine the amount of glucose present. Shake the tube gently to mix the contents before comparison.

22. Record the amount of glucose present (%) in a table.

23. *To ensure reliability of results, repeat steps 14 to 21 to obtain a total of three readings (triplicates**)** at this citrate concentration using fresh samples prepared in Part 2.

24. *To show that the glucose used is affected by citrate, a control is set up with 0 mM citrate. (The control is subject to the same environmental factors as that for the experiment.) Steps 14 to 21 are performed with equivalent volume of distilled water in place of citrate solution.

25. Repeat steps 14 to 22 using the other citrate concentrations as prepared in Part 2.

26. *To ensure reproducibility of data, repeat the entire experiment twice using freshly prepared reagents and solutions and glucose standards.

**considerations for accuracy of results
*considerations for reliability & reproducibility

D: DATA MANIPULATION AND EXPECTED RESULTS

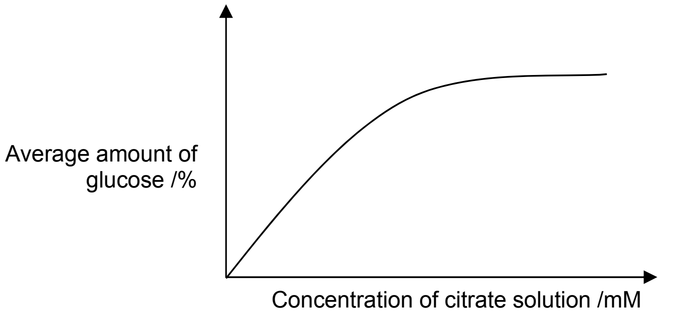
1. [*Draw Table of results*]

Table showing the amount of glucose /% at different concentrations of citrate / mM

Concentration of citrate /mM	Amount of glucose / %			
	Reading 1	Reading 2	Reading 3	Ave
0.0				
1.5				
3.0				
4.5				
6.0				
7.5				
9.0				

2. [*Draw a graph to show expected trends and/or results*]

Graph of average amount of glucose /% against concentration of citrate /mM



***3.* [*Describe and Explain the expected trends and/or results*]**

- As the citrate concentration increases, the average amount of glucose remaining increases, maximum amount being about 5%.

- This is because the amount of glucose consumed/used decreases as the concentration of citrate increases, since citrate acts as an end-product inhibitor to phosphofructokinase.
  Therefore the rate of respiration decreases as phosphofructokinase is unable to catalyse its phosphorylation reaction.

  (above-mentioned points can be written in Rationale)

SAFETY PRECAUTIONS

1. Wear safety goggles and gloves when handling chemical reagents which may be irritants (e.g. Benedict's solution, citrate solution) to the eyes and/or skin. Immediately flush the eye / wash with water when the chemical reagent comes into contact with the eyes and/or skin. If irritation persists, call for medical help.

2. Use a rag when handling the containers with hot water to prevent burns.

5 Free-response question

Write your answers to this question on the separate answer paper provided.

Your answers:
- should be illustrated by large, clearly labelled diagrams where appropriate,
- must be in continuous prose, where appropriate,
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

(a) Explain the advantages and disadvantages of the Polymerase Chain Reaction. [6]

ADVANTAGES OF PCR

1. **A cell-free method of DNA replication, do not need to remove unwanted cellular debris and vector DNA.**

2. **A specific process that targets only the desired DNA region to be copied.**

3. **PCR is accurate enough to allow small amounts of DNA to be amplified for DNA analysis,.**
   - ***E.g. the error rate of Taq polymerase is low enough to be used to identify a different allele of a gene from PCR and gel electrophoresis,***

4. **Very sensitive; a sequence from a single DNA molecule can be amplified a million-fold to form a large number of DNA molecules.**

5. **Fast and easy to use; a typical PCR run can be completed in less than two hours (cloning into a vector requires several days) and the process is automated.**

   Flexible: many different applications

DISADVANTAGES OF PCR

1. **All lengths of DNA will be multiplied BUT only a short sequence can be accurately amplified; works best up to a few hundred base pairs only. Further increase in length of target sequence decreases efficiency of amplification.**

2. **Information regarding the sequence of the DNA fragment is required for designing primer for successful amplification. If the sequences flanking the DNA segment of interest are unknown, no proper primers can be synthesized to amplify the target DNA sequence.**

3. **Samples can be easily contaminated with DNA from other sources which result in inaccurate amplification.**

4. **PCR products made by *Taq* polymerase cannot be used for cloning into bacteria because the error rate of the enzyme is too high.**

   Note : DNA strands produced by PCR are considered accurate enough for DNA analysis but not for cloning purposes.

(b) Explain how gel electrophoresis is able to show that a particular short tandem repeat (STR) has different lengths in different people. [6]

BASIS OF STR ANALYSIS

1. **STRs are repeated sequences located next to each other, called tandem repeats found in non-coding regions (introns).**
2. **As the STRs are located at the same loci, they will be found between the same 2 restriction sites.**
3. **Carrying out a restriction digest (using the same restriction enzyme) with DNA of different individuals,**
4. **would thus result in DNA fragments of different lengths, based on the number of repeats present in their STRs.**
5. **For e.g. STRs with more repeats will have longer fragments and vice versa.**
6. **Gel electrophoresis separates DNA based on their size/length.**

HOW

7. **DNA fragments, loading dye and glycerol loaded in wells at one end of the agarose gel. These wells should be near the negatively charged electrode (cathode).**

8. **Electrodes attached to both ends of the electrophoresis chamber and a direct current is applied.**

What causes DNA to move?

9. **As DNA is negatively charged due to the phosphate groups on their sugar-phosphate backbone,**
10. **under the influence of an electric field, DNA molecules will move towards the positive end /anode of the field.**

What is the basis of DNA separation?

11. **However, DNA molecules encounter resistance from the cross-linked matrix of the agarose gel when moving towards the anode**
12. **The larger the molecule, the more resistance it encounters**

(c) Discuss the social and ethical issues of the Human Genome Project. [8]

Social and Ethical Issues:

~~1~~a. (Clinical issue) Evaluation and regulation of genetic tests

1. Genetic tests are used as tools to detect gene variants associated with a specific disease or condition, as well as for paternity testing and forensics.
2. However there are issues regarding proper evaluation and regulation of genetic tests for accuracy and reliability of detection and diagnosis of diseases.
~~1.~~ There are also concerns if standards and quality-control measures are implemented in genetic testing procedures.
3.

4. Many genetic tests are marketed directly to consumers through the internet. This raises concerns because it leads to instances where customers receive clinically significant test results without appropriate counseling from a healthcare professional, and are left to interpret complex results and diagnoses on their own.
~~see 23andme.com)~~

~~2~~b. (Clinical issue) Informed consent

1. Proper counseling by certified medical professionals given to patients about the risks and limitations of genetic technology, whether it is genetic tests or gene therapy.
~~1.~~2. Proper counselling by certified medical professionals given to parents about the use of genetic information in reproductive decision making and reproductive rights.

Pts 1 and 2, so long as "proper counselling by certified medical professionals" mentioned once (either in pt 1 or pt 2)

3. Therefore, there is a need for doctors and other healthcare professionals to be trained in the dissemination of genetic information and patient counselling in order to raise the standards of patient care.

4. This way, patients can make an informed decision about their own healthcare and how best to proceed.

~~3~~c. Uncertainty issues associated with disease susceptibilities and complex conditions identified by genetic testing

1. There is concern if genetic testing should be carried out for conditions linked to multiple genes / gene-environment interactions, e.g. heart disease, for which there is currently no treatment or cure.
2. The knowledge of one's condition may have a negative impact on one's psychological / emotional well-being.
~~1.~~3. May lead to genetic discrimination

~~2.~~ d. Reproductive issue

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~~2.~~1. There is also the concern if genetic testing of an unborn fetus for such conditions should be performed when the results might lead its parents to abort the pregnancy.

e. ~~5.~~ (Social issue) Genetic discrimination / stigmatization

~~2.~~1. With genetic information from HGP, there is a concern that there will be an emergence of a group of socially marginalized individuals that are deemed to be "genetically inferior".

~~1.~~2. Discrimination by insurance companies. Genetic information may be used by insurance companies to deny or limit insurance coverage for individuals who are genetically pre-disposed to future disease onset.

Discrimination by employers:

~~1.~~ Employers may choose to hire or retain only individuals who are not genetically pre-disposed to future disease onset as they are perceived to be more productive.

3.

~~2.~~ Employers can also turn away individuals who possess genes associated with certain behavioural traits (manic depression, obsessive-compulsive disorder, bipolar disorder etc.).

~~3.~~

4.

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